X-ray applications and CT

When does a projection answer the question?

- **Chest radiography:** one projection can show the overall lung pattern and the position of tubes and lines.
- **Mammography:** a projection can show fine calcifications when there is enough contrast and resolution.
- **Fluoroscopy:** repeated projections show motion during a procedure.

When the question is **where within the body** a feature lies, overlapping structures can make a single view ambiguous.

A projection and a slice answer different questions

Chest radiograph

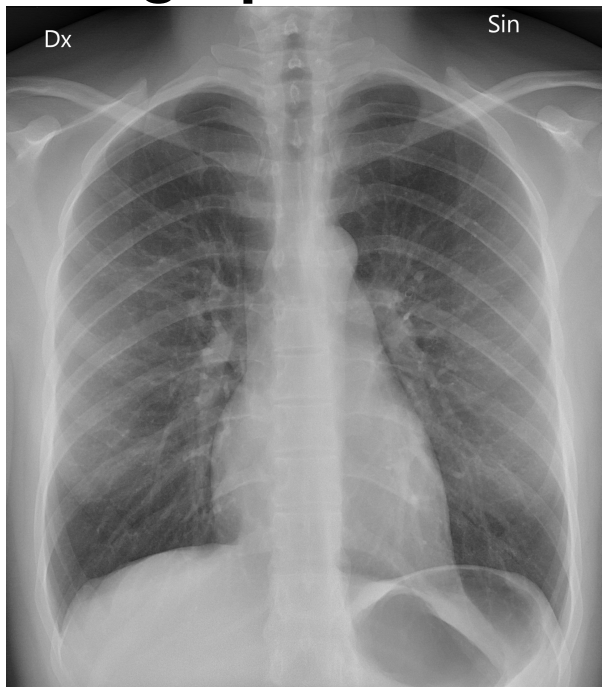


Figure 1: Normal chest radiograph

Chest CT slice

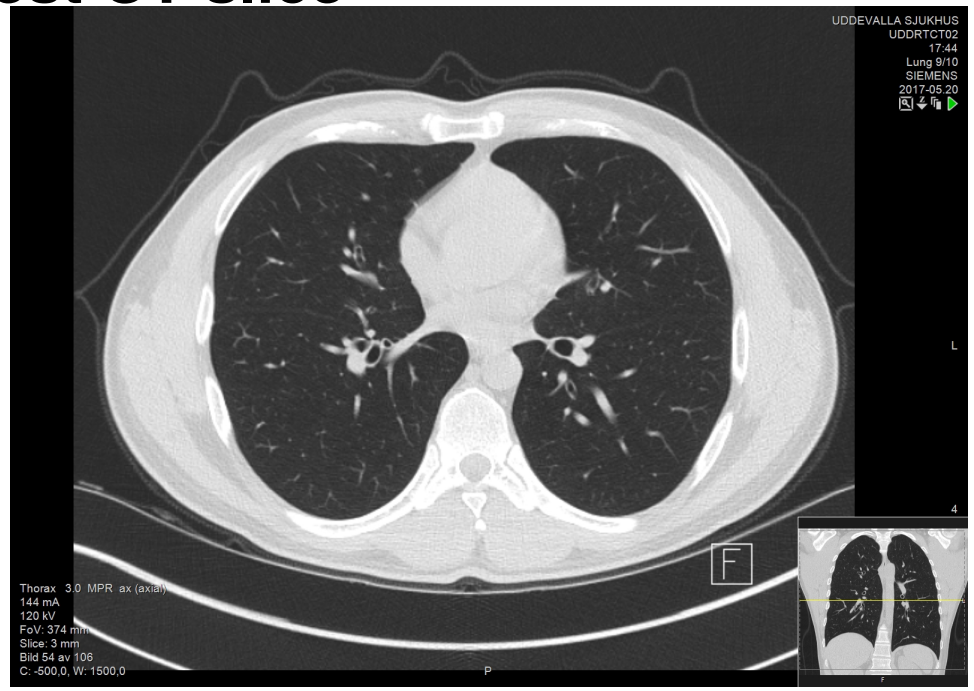


Figure 2: Axial CT slice of the chest

The projection combines structures along each ray. The slice shows their positions within one cross-section.

A CT scanner acquires views around the body

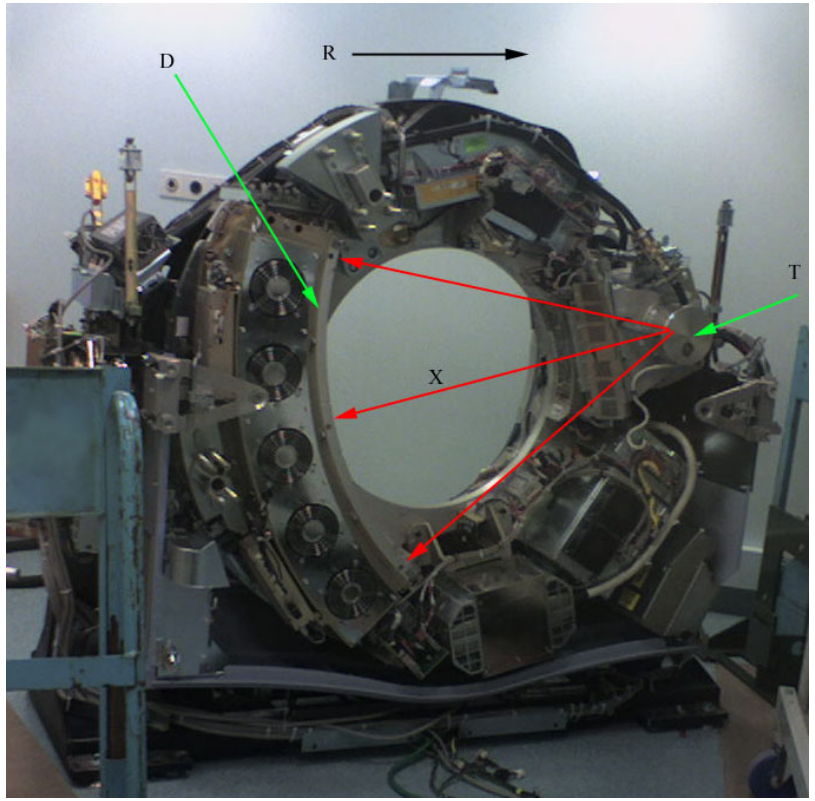


Figure 3: CT gantry with housing removed and components labeled

The **gantry** supports the X-ray tube and detector array. In this scanner they rotate together around the patient.

- **T:** tube
- **D:** detector array
- **X:** fan-shaped beam
- **R:** direction of rotation

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Each ray gives a sum of attenuation along its path

Recall the two-layer result:

$$-\ln(I/I_0) = \mu_1 x_1 + \mu_2 x_2.$$

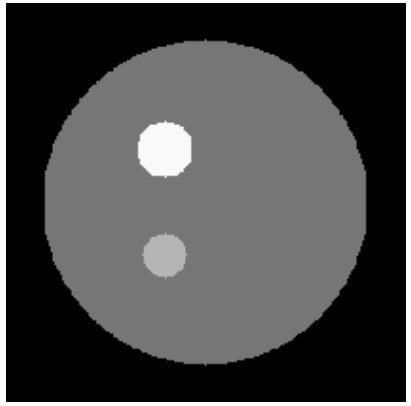
For a ray through a slice with continuously varying attenuation,

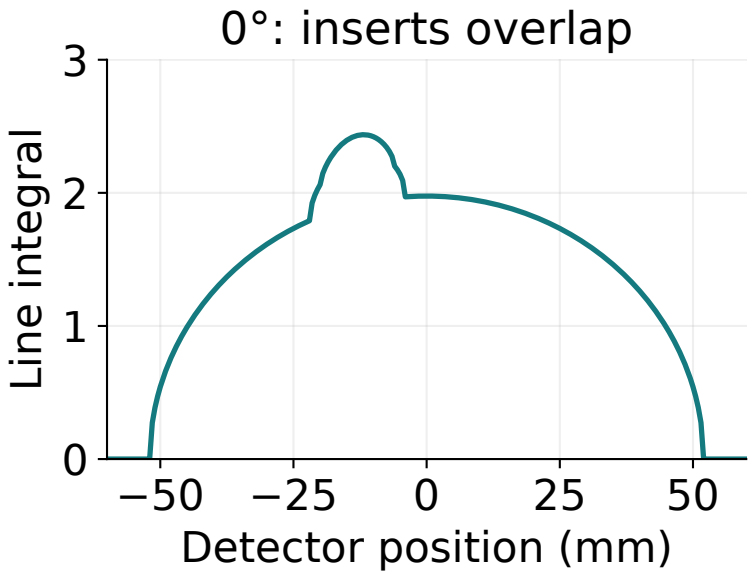
$$p = -\ln(I/I_0) = \int_{\text{ray}} \mu(x, y)\, ds.$$

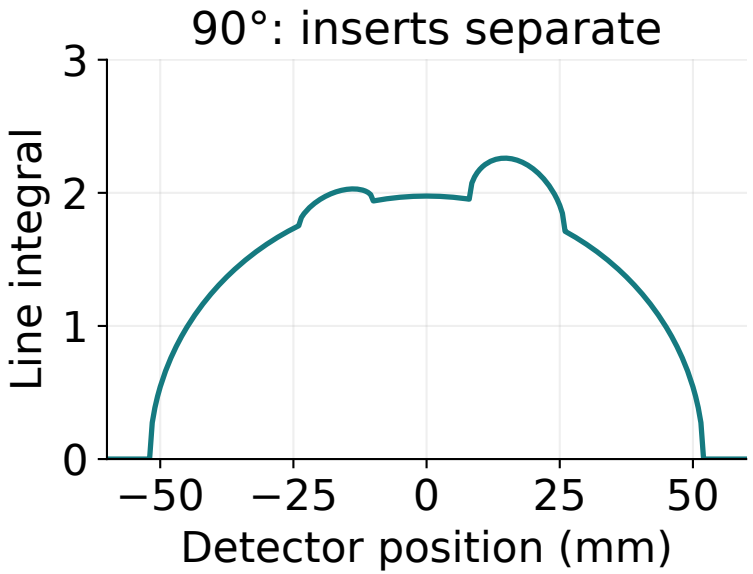
This is a **line integral**: add the local attenuation coefficient times a short distance along the ray.

One view gives a profile across the detector

Slice with two inserts

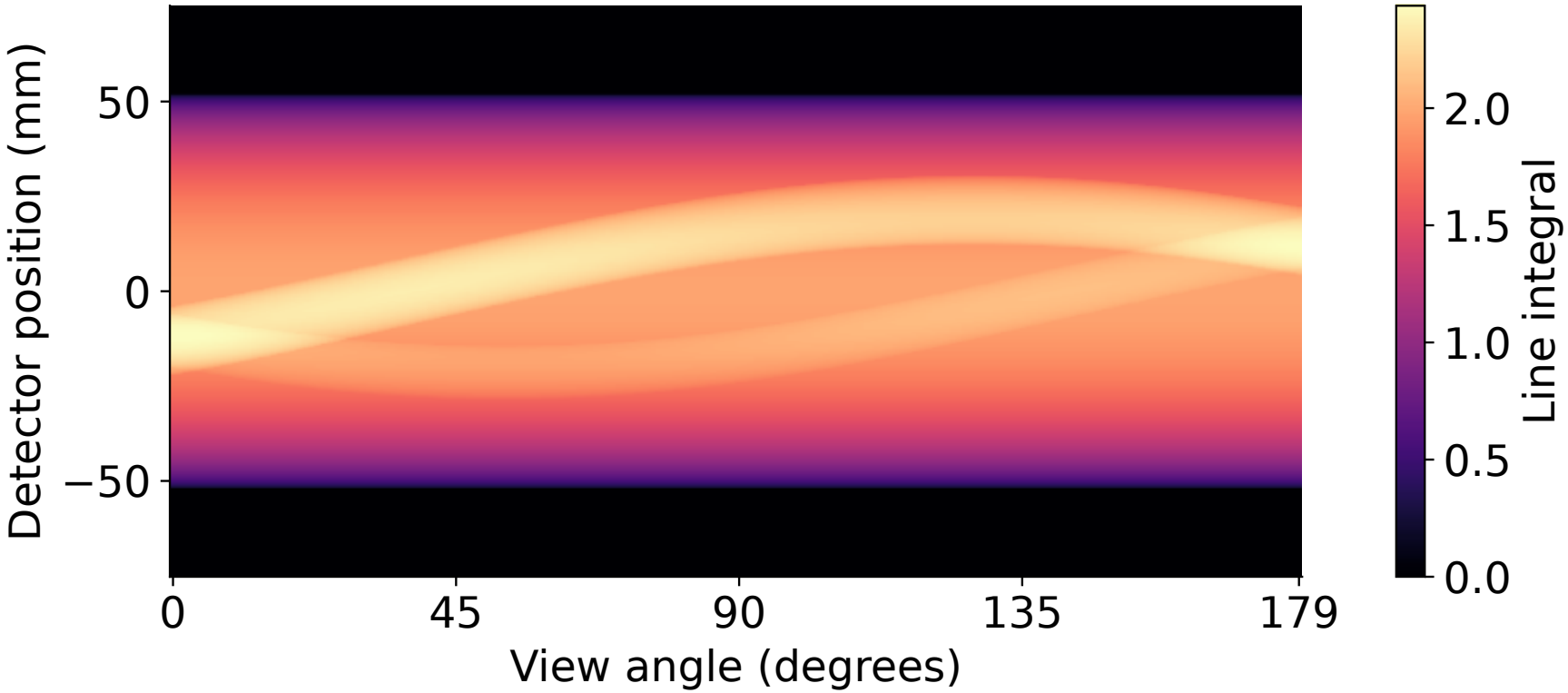






Each point on a profile is the line integral for a different ray through the same slice.

A sinogram stacks the profiles by angle



Pick an insert. How does its detector position change as the view rotates?

Backprojection spreads each value along its ray

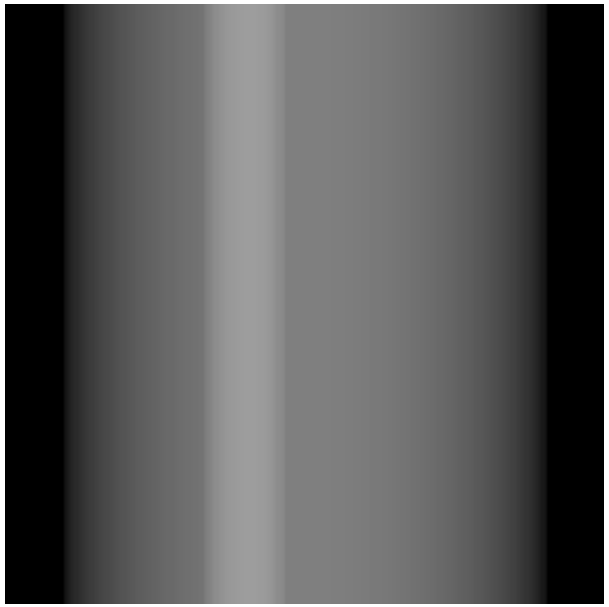
For each detector measurement, add its value back along the ray that produced it.

A single view gives stripes of possible locations. As views accumulate, rays from the same structure cross near that structure.

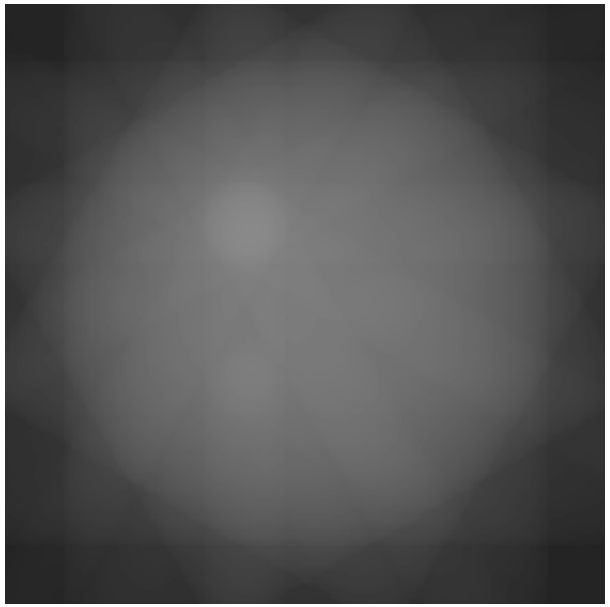
Predict: What happens to the contributions that extend beyond the object?

More views concentrate the signal

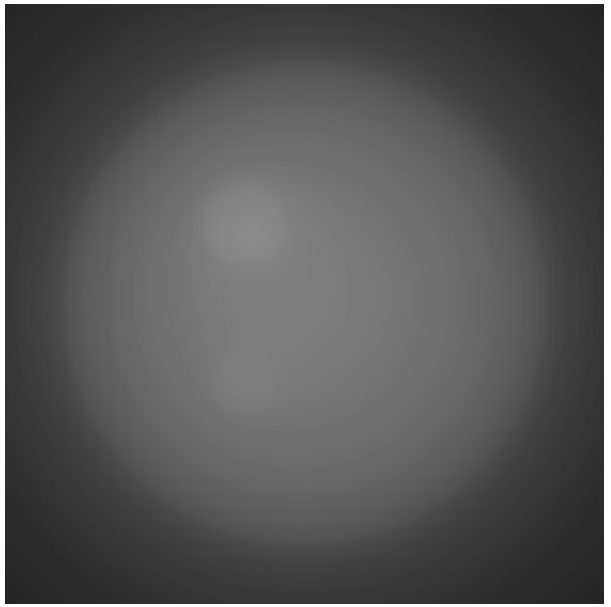
1 view



6 views



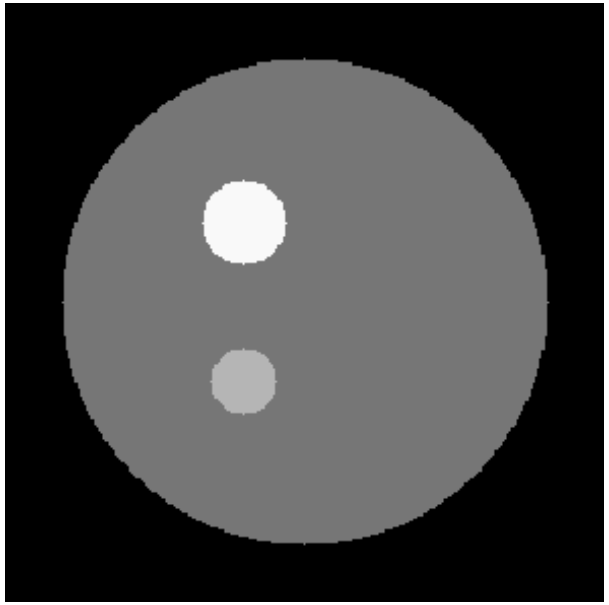
60 views



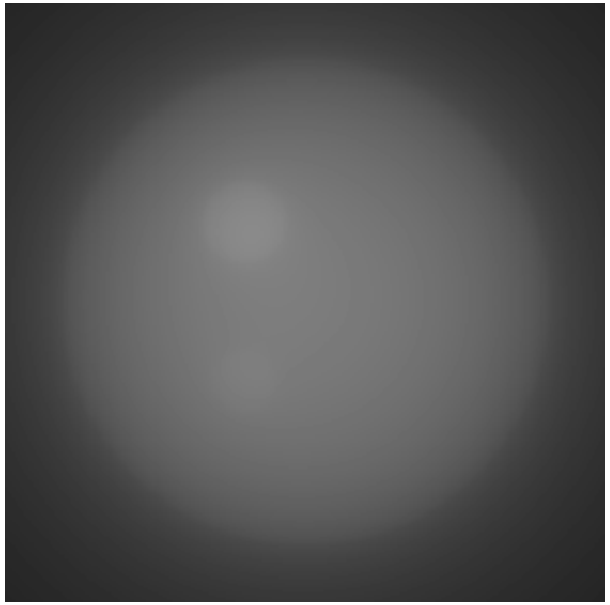
The inserts become easier to locate as views accumulate. Their edges still look broad.

Filtering sharpens the reconstructed features

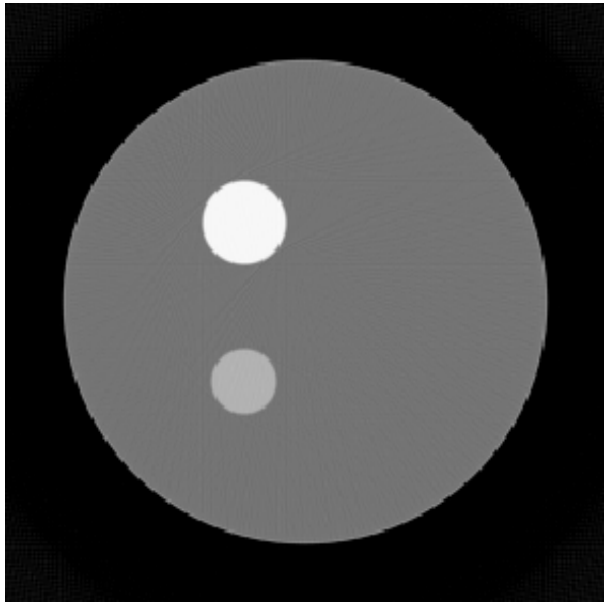
Known object



Unfiltered backprojection



Filtered backprojection

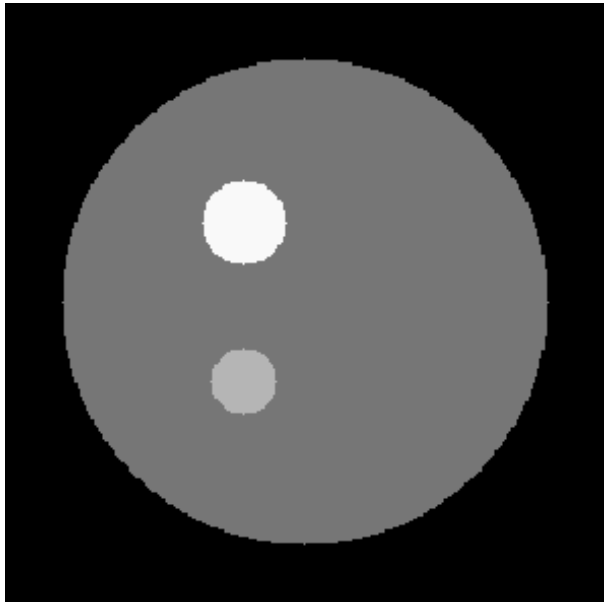


Filtered backprojection filters each detector profile before spreading its values back along the rays.

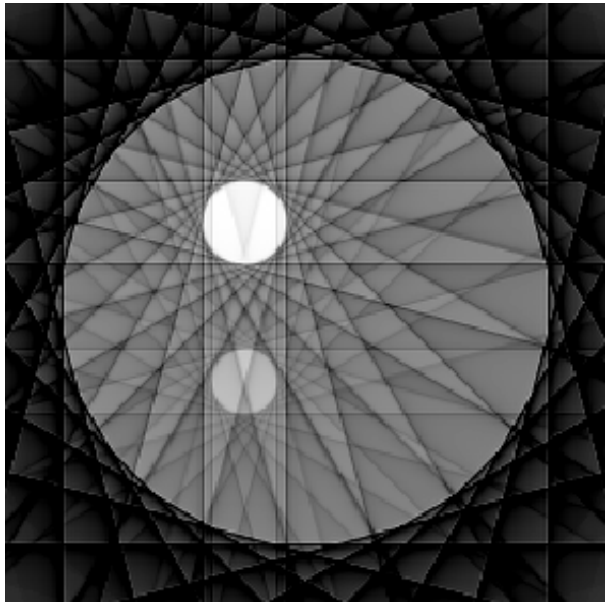
Unfiltered brightness is scaled to match the reference background.

Too few views leave streaks

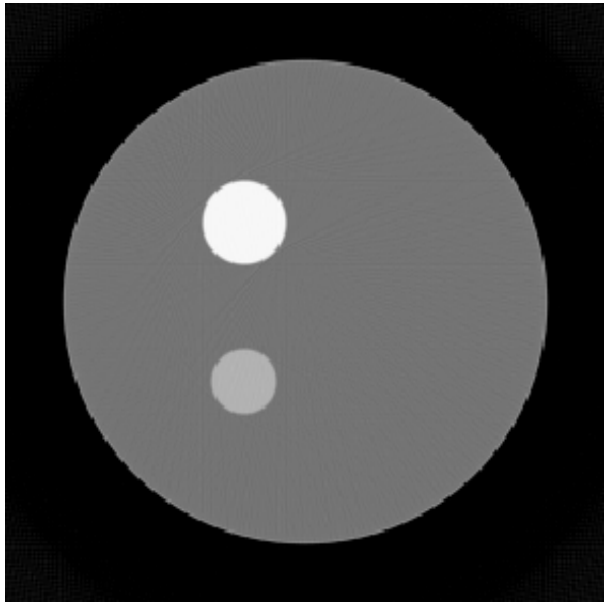
Known object



12 views



180 views



Both images use filtered backprojection. Which features in the sparse-view image are absent from the known object?

What does a CT pixel represent?

A reconstructed CT slice estimates the linear attenuation coefficient at locations within the slice.

Clinical images commonly express attenuation relative to water using **Hounsfield units (HU)**. The display maps a selected range of values to grayscale.

Exit question: What is the difference between a detector reading in one view and a pixel in a reconstructed slice?

Reading and image sources

IAEA, *Diagnostic Radiology Physics*, chapters 6 and 11.

The projection and reconstruction examples use a synthetic two-insert object and an ideal parallel-beam model.

CT gantry photograph by ChumpusRex · CC BY-SA 3.0.